Information Gathering: Interactive Methods

June 19, 2025

1 Requirement Prioritization

Systems analysts have introduced a related topic called requirements prioritization. The terminology can be misleading because the word requirements here is not the same as the information requirements we covered earlier in this chapter. In our work on information requirements, we've been engaging clients to uncover their objectives and concerns, review their organizational policies and procedures, and identify the data needed to create or refine their existing systems.
Requirements prioritization brings the client or customer into the process more directly. Simply interviewing customers or handing out questionnaires rarely elicits clear preferences about which features they value most. Asking a client which feature is the least important for their website can feel awkward and may be impossible if they haven't considered specific options in detail.
In this section, we use the terms "requirements," "features," and "deliverables" interchangeably. To illustrate each technique, imagine we have 24 requirement statements to prioritize.
Six common approaches to requirements prioritization are:

1. Simple ranking
2. 100-token method
3. MoSCoW method
4. Urgent/Important Matrix
5. Analytic hierarchy processing
6. Q-sort

These six methods are shown in Figure 1.

1.1 Simple Ranking

With simple ranking, a systems analyst asks stakeholders to order a list of requirements by assigning 1 to the most important item and 10 to the least. While this works for a handful of requirements, it becomes impractical when you need to rank 24 items.
Imagine being asked to rank every restaurant you've ever visited. After a few entries, the task feels tedious, and it's hard to distinguish whether a restaurant should be 20th or 21st on your list. Most people struggle to produce and maintain a precise ordering for long lists.

1.2 100-Token Method

In the 100-token method, the analyst places 24 glasses on a table (one per requirement) and gives each participant 100 tokens to allocate according to perceived importance. A participant could dump all tokens into one glass or spread them evenly, but such extremes effectively hand over the prioritization to others. Distributing tokens feels far more intuitive than assigning a unique rank to each requirement.

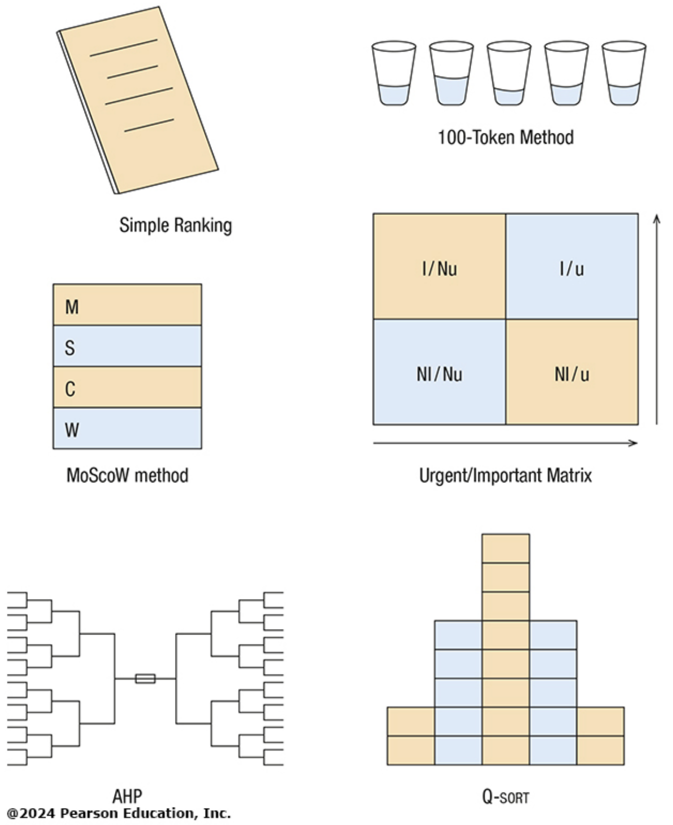

Figure 1: Six methods used for requirements prioritization

A downside is that anonymous token placement hides which department contributed which tokens, even though cross-department buy-in often predicts project success. One workaround is to use different colored tokens per department. Despite this tweak, many practitioners view the approach as overly simplistic and believe richer methods exist.

1.3 MoSCoW Method

Instead of ranking or token allocation, the MoSCoW method asks participants to sort requirements into four categories or "buckets":

1. Must
2. Should
3. Could
4. Won't

The letters M, S, C, and W serve as mnemonics for the categories. "Must" items are essential for project success. "Should" items are desirable if resources allow but are not critical. "Could" items add value once all Must and Should items are addressed. "Won't" items are explicitly excluded from consideration at this time.
Critics point out that once a requirement is labeled Won't, it's unlikely to ever resurface. Others argue that the broad categories mask finer-grained priorities within each group.

1.4 Urgent/Important Matrix

The Urgent/Important Matrix offers a grid-based alternative similar to MoSCoW. On a whiteboard, draw the horizontal axis for urgency and the vertical axis for importance. Participants place each requirement or its code in the quadrant they believe fits best.

Analysts focus first on items deemed both urgent and important by most participants. Next, they address requirements marked urgent but not important to meet looming deadlines. Tasks placed in the not-important, not-urgent quadrant are deferred indefinitely.

1.5 Analytic Hierarchy Processing (AHP)

Analytic hierarchy processing (AHP) uses pairwise comparisons to build a priority ranking. With 24 requirements, you begin by comparing them in pairs (12 matches in the first round) then continue matching winners against winners until you reach a final pairing. Repeating this tournament-style process yields a fully ranked list.
Although AHP can be time-intensive, it performs well for high-stakes, large-scale decisions. To engage participants, analysts sometimes frame the comparisons as a sports bracket, akin to the NCAA March Madness tournament.

1.6 Q-Sorts

Q-sorts resemble the 100-token and MoSCoW methods in that they involve physical cards but add a forced distribution constraint. Participants must distribute the 24 requirement cards into five piles arranged to mimic a normal distribution: 2 in the highest-priority pile, 5 in each of the next two piles, 10 in the middle pile, then 5 and 2 in the remaining piles from right (most important) to left (least important).
During sorting, participants can move cards freely. If the fifth pile already contains two cards but they believe a new card belongs there, they must shift one of the existing cards to an adjacent pile. This interactive process closely parallels the pairwise comparisons used in AHP.
Advantages of Q-sorts include:

1. Participants feel no pressure to get it right on the first pass—they can adjust as they go.
2. The forced normal distribution reduces response biases such as central tendency and leniency.
3. Analysts can identify participant groups with similar priorities by comparing their sorts, even across departments.
4. Many participants find the hands-on sorting process engaging and enjoyable.

A drawback is the need to prepare physical cards for all requirements in advance.
In summary, we have outlined six distinct approaches to requirements prioritization. Beware of opting for the easiest setup; the method that's quickest to implement may not deliver the most reliable results. It's best to understand the organization's culture first and then choose a technique that matches stakeholders' willingness to invest time and effort in ranking their requirements.